
ME 515: Carbon Materials and Technology

Course Code	ME 515
Course Name	Carbon Materials and Technology
L-T-P-C	3-0-0-3
Prerequisites	None (basic knowledge of material science expected)
Intended for	M.Tech. / M.S./ Ph.D./ B.Tech. (4th year)/ M.Sc.
Distribution	Elective
Approval	35th BoA (Aug 2020); 30th Senate (Feb 2021), in the M.Tech Materials and Energy Engineering (MEE) curriculum

Course Contents

- **Introduction to carbon**
 - Why should one study carbon materials and manufacturing?
 - Carbon economy
 - Atomic structure and hybridization
 - Carbon allotropes
 - Nomenclature and terminology [6 Lectures]
- **Microstructure of carbon materials**
 - a Bulk industrial carbon
 - Graphite: natural and pyrolytic
 - Activated carbon
 - Glass-like carbon
 - Granular amorphous carbon
 - b Carbon fibers and composites
 - Activated carbon fiber
 - Carbon fiber reinforced plastics
 - c Carbon nanomaterials
 - Graphene
 - Carbon Nanotube
 - Fullerene
 - Graphite whiskers
 - Diamond-like carbon
 - d Characterization of carbon materials (discussed in a-c) by X-Ray Diffraction, Raman spectroscopy and electron microscopy [15 Lectures]
- **Raw materials**
 - Polymer precursors (Polyacrylonitrile, cellulose, resins, PVC etc.)
 - Needle coke (petroleum and pitch based)
 - Coal and its distillation
 - Gaseous hydrocarbons (for CVD)
 - Carbon yield and mechanism of carbonization (thermodynamic and kinetic aspects) [4 Lectures]
- **Properties of carbon materials**

- Crystallinity and electrical conductivity
- Electrochemistry and surface chemistry
- Mechanical and thermal properties [4 Lectures]
- **Manufacturing techniques for carbon materials**
 - Pyrolysis (activated carbon, glassy carbon, pyrolytic graphite)
 - Electrospinning (carbon fibers)
 - Chemical vapor deposition (graphene and CNT)
 - Composite preparation (material/ binder interface)
 - Mechanical property testing methods [4 Lectures]
- **Applications**
 - Graphite electrodes
 - Carbon-fiber composites in automobile industry
 - Carbon-based micro and nano devices (sensors, microelectrodes etc.)
 - Filters and adsorbers [6 Lectures]
- **Special topics**
 - Health and environmental safety of carbon nanomaterials
 - Carbon-based flexible electronics
 - Future of carbon technology [3 Lectures]

Text Books

1. Timothy D. Burchell, **Carbon Materials for Advanced Technologies**, Elsevier, 1999.

References

1. Jenkins, G. M. & Kawamura, K., **Polymeric carbons–carbon fibre, glass and char**, Cambridge University Press, 1976).
2. Marsh, H. & Rodríguez-Reinoso, F., **Activated carbon**, Elsevier, 2006.
3. Kong, L. B., **Carbon nanomaterials based on graphene nanosheets**, CRC Press, 2017.
4. Chung, D.D.L. **Carbon fiber composites**, Butterworth-Heinemann, 1994